Cell Biology

Altair-LSFM: A High-Resolution, Easy-to-Build Light-Sheet Microscope for Sub-Cellular Imaging

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eLife Assessment

This **useful** study presents Altair-LSFM, a **solid** and well-documented implementation of a light-sheet fluorescence microscope (LSFM) designed for accessibility and cost reduction. While the approach offers strengths such as the use of custom-machined baseplates and detailed assembly instructions, its overall impact is limited by the lack of live-cell imaging capabilities and the absence of a clear, quantitative comparison to existing LSFM platforms. As such, although technically competent, the broader utility and uptake of this system by the community may be limited.

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Abstract

Although several open-source, easy-to-assemble light-sheet microscope platforms already exist—such as mesoSPIM, OpenSPIM, and OpenSpin—they are primarily optimized for imaging large specimens and lack the resolution required to visualize sub-cellular organelles or cytoskeletal architectures. Conversely, commercial systems like Lattice Light-Sheet Microscopy offer improved resolution but are complex, expensive, and alignment-intensive. To address this gap, we developed Altair-LSFM, a high-resolution, open-source, sample-scanning light-sheet microscope specifically designed for sub-cellular imaging. By optimizing the optical pathway in silico, we created a custom baseplate with precisely positioned dowel pins to simplify alignment and assembly. The system integrates streamlined optoelectronics and optomechanics with seamless operation through our open-source software, navigate. Altair-LSFM achieves lateral and axial resolutions of approximately 235 nm and 350 nm, respectively, across a 266-micron field of view after deconvolution. We validate the system's capabilities by imaging sub-diffraction fluorescent nanospheres and visualizing fine structural details in mammalian cells, including microtubules, actin filaments, nuclei, and Golgi apparatus.

Introduction

Light-sheet fluorescence microscopy (LSFM) has revolutionized volumetric imaging by enabling rapid, minimally invasive 3D investigations of diverse biological specimens[1]. By illuminating the sample with a thin sheet of light from the side and capturing 2D images in a highly parallel format, LSFM dramatically reduces photobleaching and out-of-focus blur. Interestingly, the first concept of LSFM was documented as early as 1903, and nearly nine decades later, it was adapted to visualize cochlear structures[2]. However, LSFM truly entered the limelight in 2004[1], sparking the development of numerous specialized variants—each suited to unique biological tasks. Collectively, these methods have facilitated the long-term tracking of cells through embryological development[3], the mapping of brain architecture[4] and activation patterns[5], and much more, making LSFM indispensable for a wide array of dynamic, 3D imaging applications.

While these milestones highlight LSFM's transformative potential, it wasn't until the early 2010s that researchers harnessed LSFM for sub-cellular imaging[6]. Achieving this level of detail demands optimizing both resolution and sensitivity, which are fundamentally dictated by the numerical aperture (NA) of the microscope's objectives. More specifically, lateral resolution is governed by the fluorophore's emission wavelength and the NA of the detection objective, while photon collection efficiency scales with the detection objective's NA

, making high-NA objectives essential for resolving fine, less abundantly labeled structures, while maximizing signal. Axial resolution, in turn, is set by the detection objective's depth of focus and the thickness of the illumination beam. When the illumination beam is thinner than the depth of focus, it's thickness essentially becomes the axial resolution[7]. Conversely, if the illumination beam is larger than the depth of focus, then the depth of focus takes precedence, and fluorescence elicited outside this region contributes to the image as blur. Importantly, there is a trade-off between field of view, axial resolution, and NA: pushing for high axial resolution often constrains the accessible field of view, necessitating careful mechanical and optical design choices.

Bounded by these fundamental constraints, several methods have been developed to achieve sub-cellular resolution in LSFM. For example, one can illuminate the specimen with a propagation-invariant beam[6] or an optical lattice[8]. This can be done coherently, as in Lattice Light-Sheet Microscopy (LLSM)[8], or incoherently, as in Field Synthesis[9]. However, the most used optical lattice, the four-beam "square" lattice, offers little advantage over a traditional Gaussian beam[10]. Another approach, dual view inverted Selective Plane Illumination Microscopy (diSPIM), captures images from multiple orthogonal perspectives and computationally fuses them using iterative deconvolution, significantly improving axial resolution[11]. However, this method requires precise image registration and intensive computational processing. Axially Swept Light-sheet Microscopy (ASLM)[12] extends the field of view while maintaining high axial resolution but operates at lower speeds and sensitivity compared to LLSM and diSPIM, making it less suitable for fast volumetric imaging. Oblique plane microscopy (OPM) offers another alternative by imaging an obliquely launched light sheet with a non-coaxial and complex optical train, allowing for single-objective light-sheet imaging but introducing substantial alignment challenges[13]. While these techniques offer powerful solutions for sub-cellular imaging, they all require expert assembly and precise optical alignment, limiting their widespread adoption. As a result, there remains a critical need for a high-resolution, accessible LSFM system that combines cutting-edge imaging performance with ease of assembly and operation.

To address these limitations, we developed Altair-LSFM, a high-resolution, open-source light-sheet microscope that achieves sub-cellular detail while remaining accessible and easy to use. Named after the navigational star, Altair-LSFM is built upon two guiding optical principles. First, in LLSM, the sole improvement in lateral resolution comes from the use of a higher-NA detection objective, which we incorporate to maximize both resolution and photon collection efficiency. Second, when diffraction effects are fully accounted for, a tightly focused Gaussian beam achieves a beam waist and propagation length that is comparable to that of a square lattice, eliminating the need for specialized optical components while preserving high axial resolution. By leveraging these principles, Altair-LSFM delivers optical performance on par with LLSM but without the added design complexity of LLSM. To streamline assembly and ensure reproducibility, we designed the optical layout for Altair-LSFM in silico, enabling a pre-defined optical alignment with minimal degrees of freedom. A custom-machined baseplate with precisely positioned dowel pins locks optical components into place, minimizing degrees of freedom and removing the need for fine manual adjustments. Additionally, by simplifying the optomechanical design and integrating compact optoelectronics, Altair-LSFM reduces system complexity, making advanced light-sheet imaging more practical for a wider range of laboratories.

By combining high-resolution imaging with an accessible and reproducible design, Altair-LSFM addresses a critical gap in LSFM—bringing sub-cellular imaging capabilities to a broader scientific community. Its reliance on fundamental microscopy principles rather than overly complex optical systems ensures both performance and simplicity, while its modular architecture allows for straightforward assembly and operation. By eliminating the need for specialized optics and intricate alignment procedures, Altair-LSFM significantly lowers the barrier to adoption, making advanced light-sheet imaging feasible for laboratories that lack the resources or expertise to implement more complex systems. This combination of performance, accessibility, and scalability establishes Altair-LSFM as a powerful and practical solution for a wide range of laboratories.

Results

Survey of Open-Source LSFM Designs

Before designing Altair-LSFM, we first evaluated existing open-source LSFM implementations to identify common design features, constraints, and trade-offs (**Table 1**). Many systems, such as UC2[14], pLSM[15], and EduSPIM[16], were explicitly developed with cost-effectiveness in mind, relying on low-cost components and simplified designs to maximize accessibility. Others, including OpenSPIM[17,18], OpenSPIN[19], and mesoSPIM[20,21], were optimized for imaging large specimens, such as developing embryos or chemically cleared tissues. Most of these systems employed modular construction methods based on rail carriers or cage systems, which, while reducing alignment complexity compared to free-space optics, still retain degrees of freedom that can lead to misalignment and increase setup difficulty. As a result, these microscopes generally operate at low magnification and low NA, limiting their ability to resolve sub-cellular structures. Among the surveyed designs, only diSPIM[11,22] was explicitly developed for sub-cellular imaging, built with a dovetail-based system for precise optical alignment. However, diSPIM's most widely deployed configuration uses NA 0.8 objectives, which limits its photon collection efficiency and resolution. Consequently, the cell biology community lacks an open-source light-sheet microscope that combines state-of-the-art resolution with ease of assembly, robust optical alignment, and streamlined computational processing.

Design Principles of Altair-LSFM

Building on these findings, we designed Altair-LSFM to achieve performance comparable to LLSM[8] while maintaining a compact footprint and streamlined assembly. While cost-effectiveness was considered in the design process, the inclusion of high-NA optics and performant, low-noise cameras inherently increase system cost. To simplify procurement and integration, we minimized the number of required manufacturers while maintaining high-performance components. Excluding the laser source-which varies significantly in price

Microscope	Illumination Optics	Detection Optics	Design
OpenSPIM[18]	Olympus UMPLFLN 10XW NA 0.3 (Water)	UMPLFLN 20XW NA 0.5 (Water)	Rail Carrier
X-OpenSPIM[17]	Nikon CFI Plan Fluor 10X NA 0.3 (Water)	Nikon CFI Apochromat NIR 40X W NA 0.8 (Water)	Rail Carrier
EduSPIM[16]	Zeiss LSFM 5X NA 0.1 (Air)	Zeiss LD Epiplan 5X NA 0.13 (Air)	Cage System
OpenSPIN[19]	Nikon CFI Plan Fluor 4X NA 0.13 (Air) Nikon Plan Fluor 10X NA 0.3 (Air)	Nikon CFI Plan Fluor 4X NA 0.13 (Air) Nikon CFI75 LWD 16X NA 0.8 (Water)	Rail Carrier
UC2[14]	Generic 4X NA 0.14 (Air)	Generic 4X NA 0.14 (Air) Generic 10X NA 0.3 (Air)	3D Printed Components
pLSM[15]	Mitutoyo Plan Apo 10X NA 0.28 (Air)	Mitutoyo Plan Apo 10X NA 0.28 (Air) ASI 54-10-12 16.67X NA 0.4 (Multi-Immersion)	Cage System
descSPIM[28]	500- and 150-mm cylindrical lenses (Air)	Thorlabs TL2X-SAP 2X NA 0.1 (Air)	Cage System
mesoSPIM[21]	Nikon 50 mm f/1.4 G (Air)	Olympus MVPLAPO 1X NA 0.15 (Air)	Rail Carrier
BT-mesoSPIM[20]	Nikon 50 mm f/1.4 G (Air)	Variable. Magnification 2-20X, NA 0.1-0.28 (Air)	Cage System
diSPIM[11]	Nikon CFI Apochromat NIR 40X W NA 0.8 (Water)	Nikon CFI Apochromat NIR 40X W NA 0.8 (Water)	Dovetail Tube System
CompassLSM[29]	Olympus XLFLUOR 4x NA 0.28 (Air/Water)	Olympus MVX PLAPO 1x NA 0.25 (Air) Olympus MVX PLAPO 2XC NA 0.5 (Air) Olympus UPlanFL 4x NA 0.13 (Air) Nikon CFI Plan Apo 10XC NA 0.5 (Water)	Rail Carrier
Lattice Light-Sheet Microscopy (LLSM)[8]	Special Optics 54-10-7 28.6X NA 0.67 (Water)	Nikon CFI75 Apochromat 25XC NA 1.1 (Water)	Free Space Optics
Altair-LSFM	Thorlabs TL20X-MPL 20X NA 0.6 (Water)	Nikon CFI75 Apochromat 25XC NA 1.1 (Water)	Custom Baseplate & Dovetail Tube System

Table 1.

LSFM variants and their associated illumination and detection optics.

The table lists the type of microscope, the illumination and detection optics-including NA where available and immersion type in parentheses-as well as the overall design architecture (e.g., rail carrier, cage system, etc.).

depending upon the number of lasers, power, and modulation requirements, the estimated price for Altair-LSFM is $150,000. A detailed list of all system components, their sources, and associated costs is provided in **Supplementary Tables S2** and **S3**.

Altair-LSFM is configured to operate in a sample-scanning format with a detection path that is nearly identical to that of LLSM, ensuring similar lateral resolution (~230 nm × 230 nm × 370 nm) and photon collection efficiency. Specifically, we paired a 25x NA 1.1 water-dipping physiology objective (Nikon N25X-APO-MP) with a 400 mm achromatic tube lens (Applied Scientific Instrumentation), ensuring Nyquist sampling (130 nm pixel size) across the full width of a standard 25 mm CMOS camera (Hamamatsu Orca Flash 4.0 v3), yielding a total field of view of 266 microns. Emission filters were positioned in the focusing space immediately before the camera, and the entire detection assembly was built around a dovetail-based tube system, to ensure robust alignment and mechanical stability (**Figure 1a**). To facilitate precise axial positioning and accommodate different sample types, the entire detection assembly was mounted on a 50 mm travel focusing stage.

With the detection path establishing the necessary criteria for resolution, field of view, and optical alignment (e.g., beam height), we next designed the illumination system. In LSFM, the foci of the illumination and detection objectives must precisely overlap without mechanical interference, limiting the choice of compatible optics. To meet these requirements, we selected a 20x NA 0.6 long-working distance water immersion objective (Thorlabs TL20X-MPL). While Special Optics' equivalent objective offers a slightly higher NA, the Thorlabs objective allows for imaging on coverslips larger than 5 mm in diameter, improving ease of handling for end users. Guided by these constraints, we selected optical components capable of generating a theoretically diffraction-limited light-sheet using straightforward magnification calculations. To further improve illumination quality and reduce shadowing artifacts, we included a resonant multidirectional beam pivoting mechanism, ensuring even excitation throughout the sample[23]. The complete illumination system was designed for an input beam with a 2 mm diameter, which first passes through an achromatic doublet lens (f = 30 mm) followed by a second achromatic doublet (f = 80 mm), which expands and collimates it before reaching an achromatic cylindrical lens (f = 75 mm). This cylindrical lens focuses the beam in one direction, forming the initial light-sheet. The focal length of the cylindrical lens was chosen to be sufficiently large (≥ 60 mm) to allow adequate spacing between optical elements in the physical implementation of the system. The shaped beam is then directed onto a resonant galvanometer (**Supplementary Figure 1**). After reflection from a 45-degree tilted mirror, the beam is relayed through an achromatic doublet (f = 250 mm) before entering the back aperture of the illumination objective, where it is finally focused onto the sample. This optical arrangement ensures a well-defined, dynamically pivoted light-sheet that provides uniform illumination while mitigating shadowing effects.

In Silico Optimization of Altair-LSFM

To ensure optimal illumination performance, we modeled the full illumination pathway of Altair-LSFM in Zemax OpticStudio (Ansys), systematically optimizing the relative placement of every optical element to achieve the desired focusing and collimation properties (**Figure 1b**). Each lens was iteratively adjusted to minimize aberrations and ensure precise beam shaping, enabling the formation of a well-defined light-sheet. The design was centered around a 488 nm illumination wavelength, with spatial axes defined following standard conventions: the Y-dimension represents the laser propagation direction, Z corresponds to the detection axis, and X is orthogonal to both. The final illumination system, depicted in **Figure 1b**, was optimized to generate a diffraction-limited light-sheet with a full-width-half-maximum (FWHM) of ~0.385 μm in Z, spanning the full 266 μm Field of view, as shown in **Figures 1c-d**.

Beyond idealized modeling, designing a physically realizable system requires an understanding of how fabrication tolerances affect optical performance. To assess system robustness, we performed a tolerance analysis, which quantifies sensitivity to mechanical perturbations. This analysis

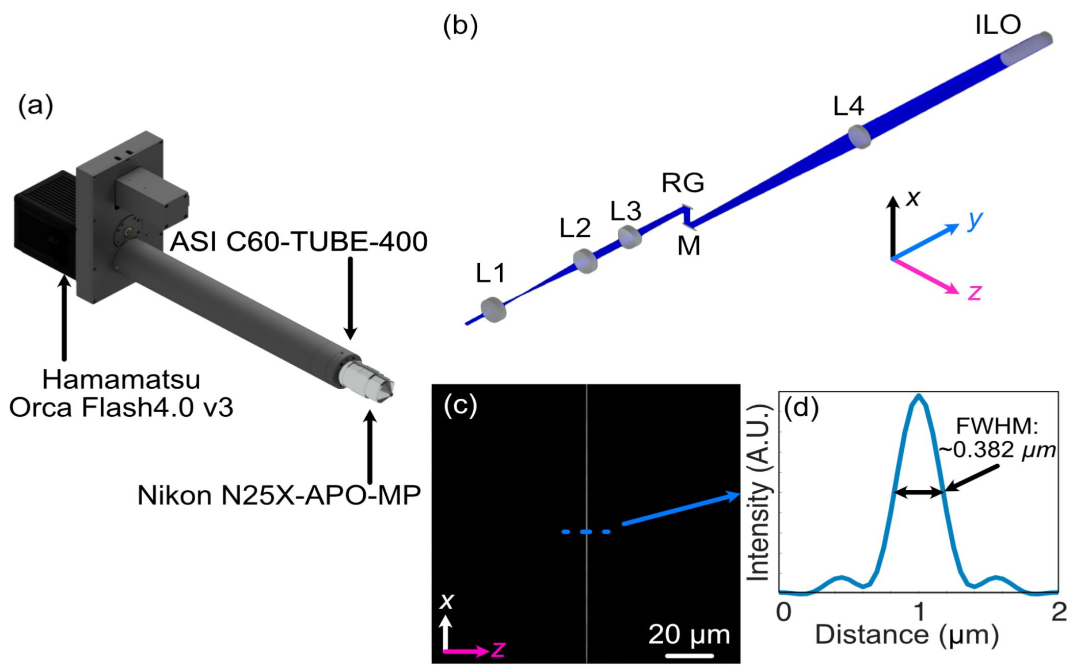


Figure 1.

(a) Rendering of the detection arm elements. **(b)** Zemax Opticstudio layout and beam path of optimized illumination arm, where L1 is an f = 30 achromatic doublet, L2 is an f = 80 mm achromatic doublet, L3 is the f = 75 mm achromatic cylindrical doublet, and ILO is our TL20X-MPL illumination objective. **(c)** The simulated light-sheet beam profile in the xz plane at the focus of the illumination objective. **(d)** The cross-sectional profile through the center of the light-sheet beam profile in (c), where the FWHM of the light-sheet was found to be 0.382 μm.

evaluates how small positional or angular deviations of optical elements—caused by manufacturing imperfections—impact key performance metrics such as light-sheet thickness and displacement from the ideal position, allowing us to systematically evaluate system stability (**Supplementary Figure 2a**). The perturbations analyzed were based on standard machining tolerances, typically ±0.005 inches, with higher-precision machining achievable at ±0.002 inches at increased cost. Given that Altair-LSFM was designed assuming the use of Polaris mounts, which incorporate DIN-7m6 ground dowel pins to aid with alignment, we considered the impact of angular misalignments caused by dowel pin positioning errors. In the worst-case scenario—where one dowel pin was offset by +0.005 inches and the other by -0.005 inches—the resulting angular deviation was ~1.45 degrees (**Supplementary Figure 2b**). To further assess system resilience, we conducted Monte Carlo simulations incorporating these perturbations, simulating a range of misalignment scenarios to quantify their effect on light-sheet performance. Our results showed that finer machining tolerances resulted in a worst-case performance closer to the nominal system, as visualized in **Supplementary Figure 2c**, which compares the nominal, best, and worst configurations. Notably, the analysis identified that angular offsets in the galvo mirror had the most significant impact on light-sheet quality, highlighting the importance of tighter machining tolerances for this component to maximize system stability and performance.

Optomechanical Design of Altair-LSFM

Based on our simulation results, standard machining tolerances were deemed sufficient to construct a custom baseplate that ensures robust alignment, repeatable assembly, and a compact, plug-and-play design. Unlike cage-or rail-based systems, custom baseplates minimize variability by enforcing a fixed spatial relationship between optical components, enabling assembly by non-experts. Where possible, we eliminated all unnecessary degrees of freedom, restricting manual adjustments to only a few critical components. Specifically, we retained laser collimation (tip/tilt/axial position), galvo rotation, folding mirror alignment (tip/tilt), and objective positioning (tip/tilt/axial position).

To translate the numerically optimized positions of each optical element into a manufacturable design, the coordinates were imported into computer-aided design (CAD) software (Autodesk Inventor), ensuring precise positioning of all associated optomechanics. Where possible, Polaris optical posts and mounts were used to maintain consistency in mounting schemes and element heights. For components where a commercially available Polaris-compatible mount did not exist, such as the rotation mount (Thorlabs RSP1) for the cylindrical lens and the horizontal aperture (Thorlabs VA100), custom adapters were developed to seamlessly integrate them into the system. This approach allowed us to account for the offset between the optical element and its mechanical mount, ensuring that the baseplate was precisely machined with dowel pin locations and mounting holes (**Figure 2a**, **Supplementary Figure 3**). Additionally, the baseplate features four mounting holes at its corners, spaced such that it can be directly secured to an optical table or elevated using additional posts, allowing for easy adjustment of the illumination path height. This modular, precision-engineered design ensures both ease of use and long-term mechanical stability, enabling integration of Altair-LSFM into alternative experimental setups.

Beyond the illumination path, the full microscope system incorporates a variety of translational and custom mounting elements to facilitate precise sample positioning and stable imaging. A sample chamber was designed and 3D printed to match the working distances and clearances of the chosen illumination and detection objectives, offering two port configurations: one for traditional orthogonal imaging and another linear configuration that allows direct imaging of the light-sheet itself (**Figure 2b**, **Supplementary Figure 4**). Each port is equipped with two sets of O-rings, creating a liquid-proof seal around the objectives while still permitting smooth translation of the detection objective for focusing. Additionally, the snug fit of the O-rings naturally guide the user toward proper positioning of the illumination and detection objectives, decreasing the likelihood of alignment errors. The sample positioning assembly consists of three motorized translation stages (Applied Scientific Instrumentation), enabling precise positioning of the sample

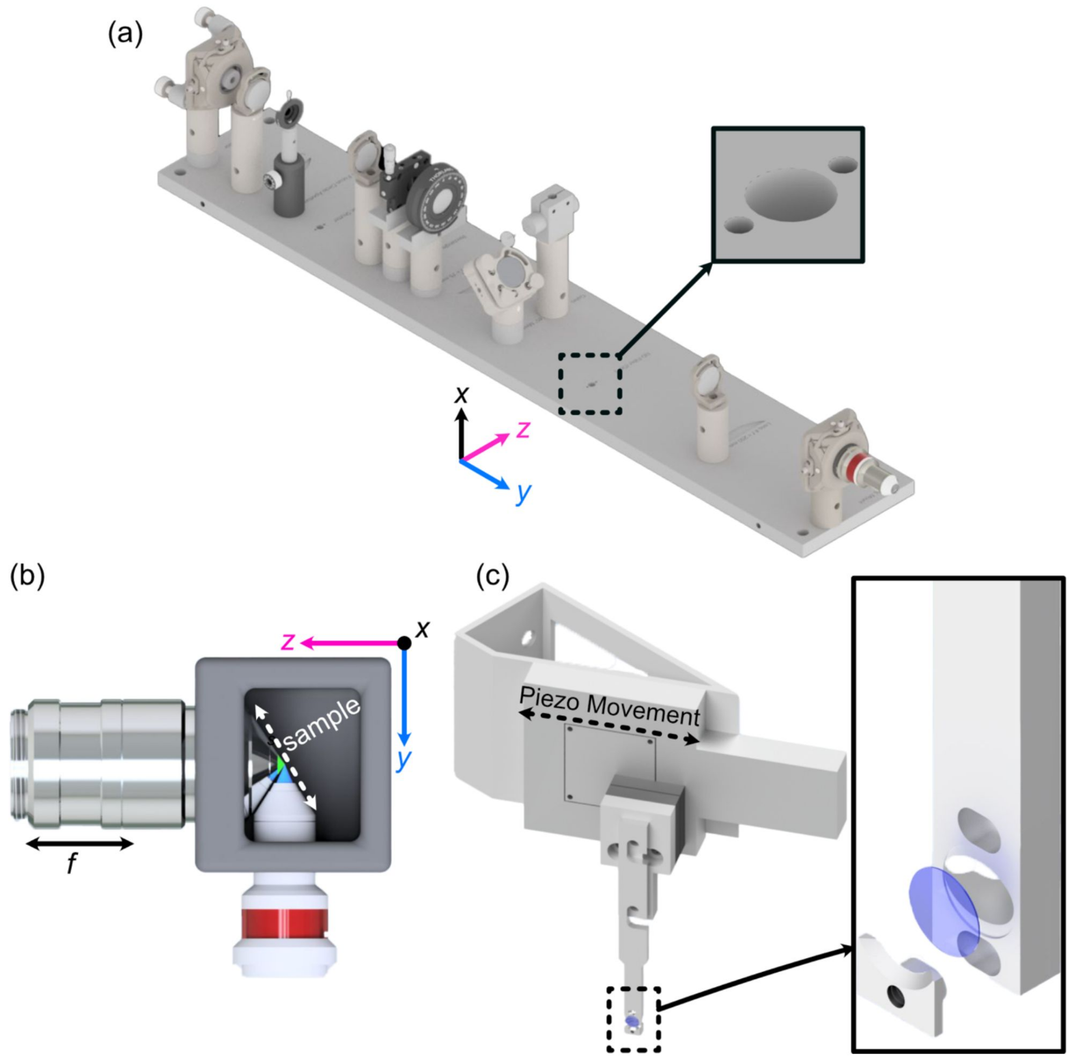


Figure 2.

(a) Rendering of the completed illumination arm baseplate, with an inset showing the dowel pin holes compatible with the Polaris mounting line from Thorlabs. **(b)** Overhead view of the imaging configuration of our system, where our detection objective and illumination objectives are placed orthogonal to each other and the sample is scanned diagonally in the space between them in the axial direction shown by the white dashed line. **(c)** Rendering of our sample mounting and translation system. Here, a piezo motor is mounted onto an angled adapter to allow precise translation over the diagonal region between the objectives. Our custom 5 mm coverslip sample holder is also featured, where the inset shows an exploded assembly of the holder.

in x, y, and z. To enable rapid Z-stack acquisition, we designed an angled bracket (θ = 29.5 degrees) for mounting a high-speed piezo (PiezoConcept HS1.100), which attaches directly to the sample positioning stages (**Figure 2c**). The sample is secured using a custom-designed sample holder for 5 mm glass coverslips, which features a clam-based mechanism-the coverslip is placed within a circular recess and secured in place by a screw-down clamp. Due to the angled sample scanning configuration, our collected image stacks must undergo a deskewing operation. All custom component designs, and deskewing software, are available for download at *https://thedeanlab.github.io/altair*.

Optoelectronic Design of Altair-LSFM

In addition to simplifying the optomechanical design, we also streamlined the electronics and control architecture of Altair-LSFM to minimize complexity and improve system integration. To achieve this, we consolidated all control electronics into a single controller (TG16-BASIC, Applied Scientific Instrumentation), which manages the operation of all linear translation stages (X, Y, Z, and F), as well as the power supply for the resonant galvo and sample scanning piezo. This approach significantly reduces the number of auxiliary controllers and power supplies, simplifying the physical setup. Currently, all timing operations are performed using a 32-channel analog output device (PCIe-6738, National Instruments), which is responsible for generating the global trigger, controlling the camera's external trigger, modulating the laser through analog and digital signals, setting the piezo control voltage, and providing the DC voltage for adjusting the resonant galvo amplitude. An overview of the electronics used in the system, along with an associated wiring diagram, is provided in **Supplementary Figure 5** and **Supplementary Table 1**.

Alignment and Characterization of Altair-LSFM

To evaluate optical performance, we first assembled and aligned the Altair-LSFM illumination system. The fiber-coupled laser source (Oxxius L4CC), which provides four excitation wavelengths (405 nm, 488 nm, 561 nm, and 638 nm), was introduced and collimated using tip/tilt mounts. The collimated beam was then directed onto the resonant galvo, which was rotated to reflect the beam downward toward the optical table. From there, the folding mirror was adjusted to guide the beam along the optical axis of the remaining components. Finally, the lateral position of the illumination objective was fine-tuned to ensure coaxial back-reflections, completing the alignment process. With the optical path aligned, we proceeded to validate system performance by characterizing the generated light-sheet by imaging it in transmission. As shown in **Figure 3a**, the light-sheet focus spans the full 266 μm field of view, closely matching our simulation results. Cross-sectional analysis of the full-width-at-half-maximum (FWHM) in the z-dimension, presented in **Figure 3b**, reveals a z-FWHM of ~0.415 μm. To assess the system's resolution, we imaged 100 nm fluorescent beads. The point spread function of a single isolated fluorescent bead is shown in **Figures 3c-e**, providing orthogonal perspectives of the bead's intensity distribution. The Gaussian-fitted distribution of FWHM measurements, performed on a population of fluorescent beads across a z-stack, is shown in **Figure 3f**. Prior to deconvolution, the average FWHM values measured across the bead population were 328 nm in x, 330 nm in y, and 464 nm in z. After deconvolution with PetaKit5D, these values improved to 235.5 nm (x), 233.5 nm (y), and 350.4 nm (z), achieving our desired resolution goals for sub-cellular imaging.

Sub-Cellular Imaging with Altair-LSFM

To demonstrate the imaging capabilities of Altair-LSFM in a biological context, we prepared and imaged mouse embryonic fibroblasts (MEFs) cells stained for multiple subcellular structures. The staining protocol enabled visualization of the nucleus (DAPI, 405 nm, cyan), microtubules (488 nm, gray), actin filaments (561 nm, gold), and the Golgi apparatus (638 nm, magenta), corresponding to the excitation channels of our system. Deconvolved maximum intensity projections of the labeled cells is shown in **Figure 4a** and **Figure 4f**, with each individual corresponding fluorescence

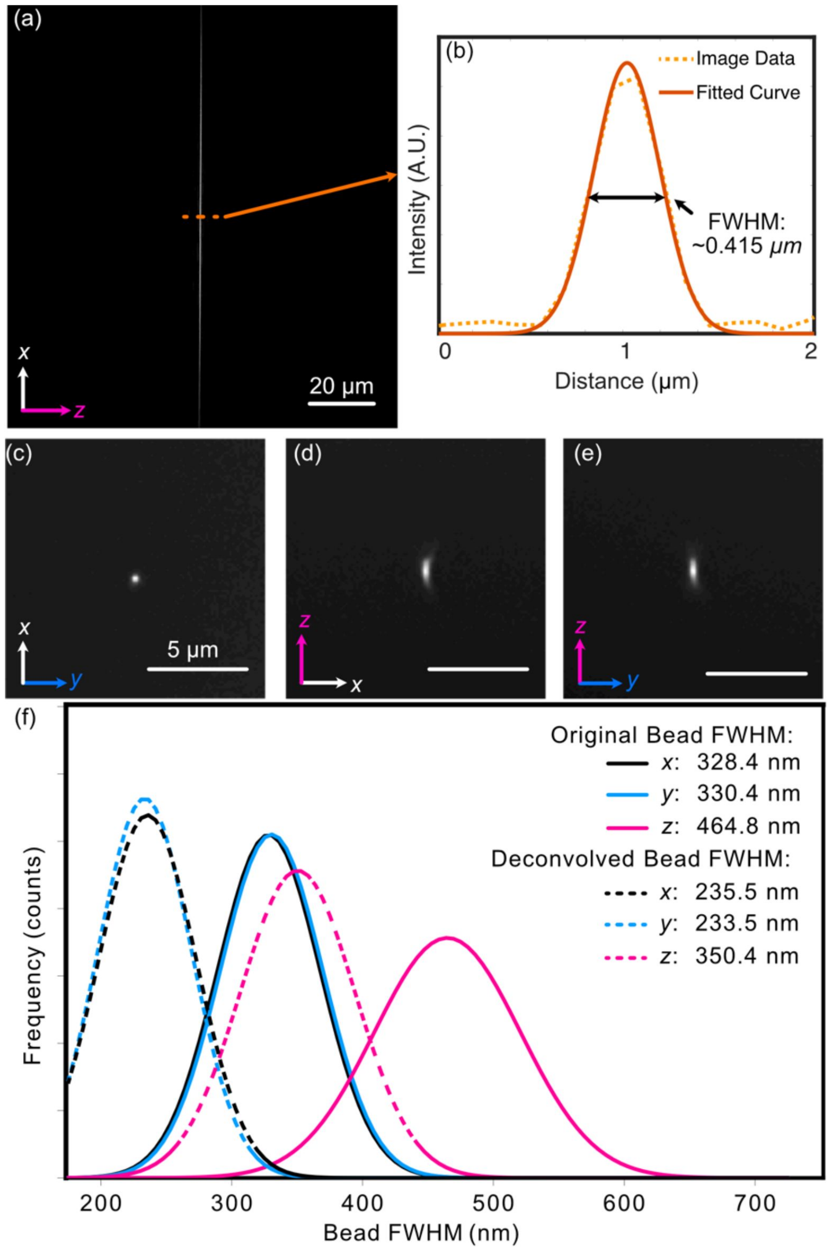


Figure 3.

(a) Experimental light-sheet beam profile at the focus **(b)** The center cross-section profile of (a), showing both raw data and a fitted curve with a FWHM of ∼0.415 µm. **(c-e)** Max intensity projections for an isolated 100 nm fluorescent bead in each of the 3 orthogonal planes **(f)** Gaussian-fitted distribution of the FWHM of beads imaged in a z-stack in each dimension both before (solid) and after (dashed) deconvolution.

channel presented in **Figures 4b-e** and **Figures 4g-j**. The imaging results reveal fine nucleolar features within the nucleus, with perinuclear Golgi structures distinctly visible. Additionally, stress fibers are clearly resolved in the actin channel, and individual microtubules appear well-defined, highlighting the system's ability to capture cytoskeletal structures with high resolution. These results confirm that Altair-LSFM provides the sub-cellular resolution, optical sectioning, and multi-color imaging performance necessary for quantitative biological imaging applications.

Discussion

In this work, we demonstrated the viability of a baseplate-based approach for the disseminating a high-performance light-sheet microscope that is both accessible and easy to assemble by non-experts. By combining optical simulations, precision-machined component design, and experimental validation, we developed Altair-LSFM, a system that delivers sub-cellular resolution imaging with minimal alignment requirements. Characterization using fluorescent beads confirmed that Altair-LSFM achieves a resolution of 328, 330, and 464 nm before deconvolution, which improves to 235.5, 233.5, and 350.4 nm after deconvolution, in X, Y, and Z, respectively. These values are on par with LLSM, confirming that our approach achieves state-of-the-art performance but in a streamlined, cost-effective, and reproducible format. A complete parts list, and estimated cost, are provided in **Supplementary Tables 2 and 3**, respectively, offering a transparent roadmap for users looking to adopt and build the system.

Building on this successful prototype, future efforts will focus on expanding both imaging capabilities and user experience. One natural evolution of this approach is the development of more advanced light-sheet microscope designs, such as Axially Swept Light-Sheet Microscopy[12,24] and Oblique Plane Microscopy[13,25], which offer additional flexibility for imaging a broader range of sample types. Another avenue for improvement is the optimization of Altair-LSFM for cleared-tissue imaging, further extending its applicability into tissue contexts. Additionally, we aim to eliminate the need for an external analog output device, consolidating all triggering and waveform generation within a single unified controller, which will reduce hardware dependencies and enhance system efficiency.

Another important consideration is the long-term scalability and routine maintenance of Altair-LSFM in a variety of lab environments. While our initial prototype has shown reliability, multi-site benchmarking and community feedback will be pivotal to ensure consistent, reproducible performance across different setups. Future enhancements—such as self-alignment routines—could further boost imaging quality and throughput. To accelerate widespread adoption, we have thoroughly documented the entire assembly process in our GitHub repository[26], which is also provided as a Supplementary Document, and are holding on-site workshops aimed at equipping researchers with the hands-on skills needed to assemble and operate a Altair-LSFM system. By fostering open collaboration and continuous software-hardware integration, we envision Altair-LSFM evolving into a robust, ever-improving platform, well-suited to meet future research challenges in high-resolution, volumetric imaging.

A persistent challenge in advanced microscopy is the lengthy interval—from instrument conceptualization to commercialization—often spans close to a decade. By offering modular, high-performance microscope designs that can be assembled and operated by non-experts, we hope to drastically shorten this timeline. Integrating these systems with navigate[27], our open-source microscope control software, will further democratize intelligent imaging workflows and broaden the reach of cutting-edge instrumentation. With high-NA imaging, reduced mechanical and optical complexity, and lower cost, Altair-LSFM stands poised to accelerate LSFM adoption, delivering a powerful yet accessible solution for researchers immediately seeking access to high-resolution, cutting-edge, volumetric imaging.

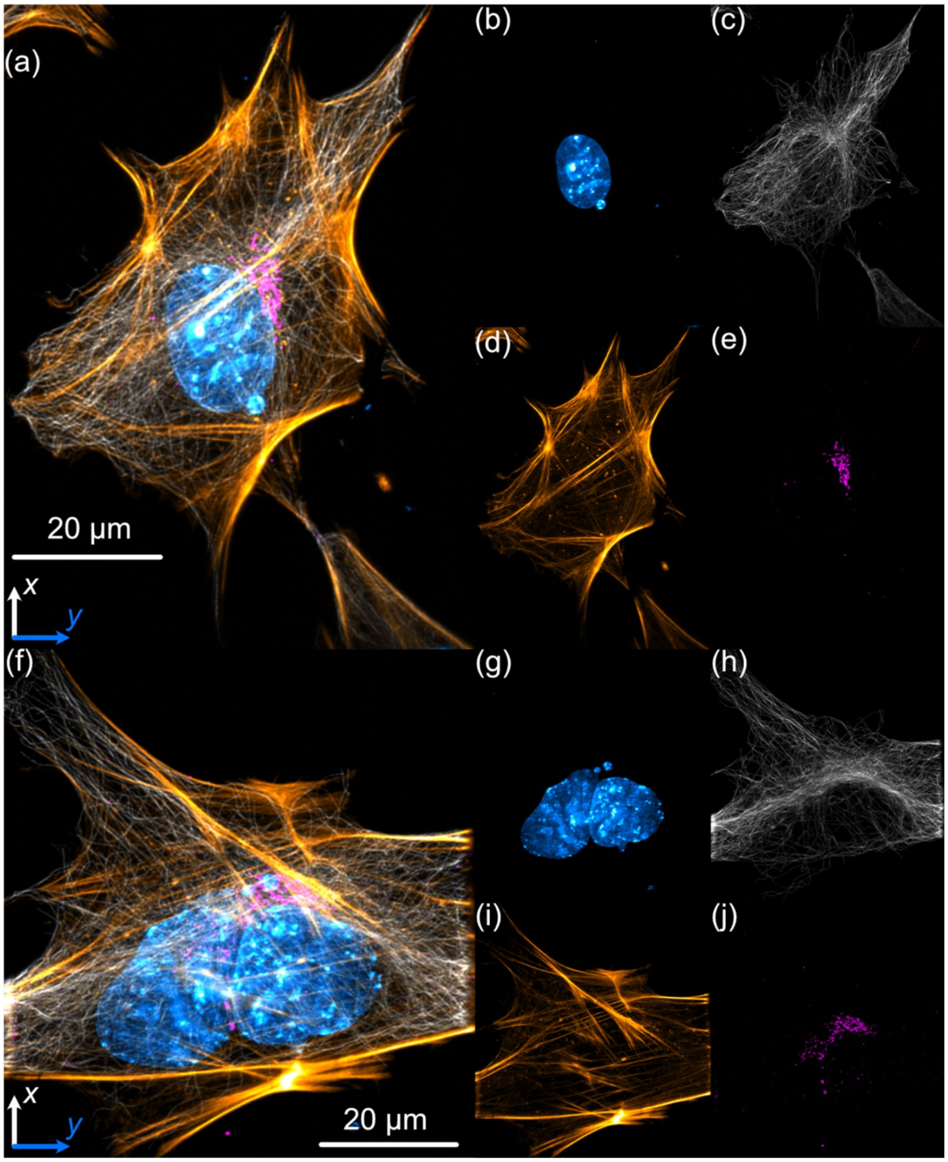


Figure 4.

Lateral maximum intensity projections of mouse embryonic fibroblasts (MEFs) fluorescently labeled with nuclei (cyan), tubulin (gray), actin (gold), and the Golgi apparatus (magenta).

(a) Maximum intensity projection of the full z-stack in the xy plane. (b–e) Individual channels corresponding to (a): (b) nuclei, (c) microtubules, (d) actin, and (e) Golgi apparatus. (f) Maximum intensity projection of a second z-stack in the xy plane. (g–j) Individual channels corresponding to (f): (g) DAPI, (h) microtubules, (i) actin, and (j) Golgi apparatus. Nuclei were labeled with DAPI, actin filaments with phalloidin, and both microtubules and the Golgi apparatus were stained using indirect immunofluorescence.

Materials and Methods

Acquisition and Simulation Computer

All microscope control and optical simulations were performed on a Colfax SX6300 workstation, configured to handle high-speed data acquisition and processing. It is equipped with dual Intel Xeon Silver 4215R processors (8 cores, 16 threads, 3.2 GHz), 256 GB of DDR4 3200 MHz ECC RAM, a 7.68 TB Samsung PM9A3 NVMe SSD for high-speed data acquisition, a 20 TB Seagate Exos X20 HDD for long-term data storage, a PNY NVIDIA T1000 4 GB GPU, and an Intel X710-T2L dual-port 10GbE.

Cell Culture and Fixation

Mouse embryonic fibroblast (MEFs) cells were cultured in Dulbecco's Modified Eagle's Medium (DMEM, Gibco) supplemented with 10% fetal bovine serum (FBS, Gibco) and 100 µg/mL penicillin-streptomycin. The cells were maintained at 37°C in a humidified incubator with 5% CO_2 and cultured in a 12-well plate on 5 mm cover slips, pre-rinsed with 70% ethanol. After reaching ∼50% confluency, cells were rinsed with 1x Phosphate-Buffered Saline (PBS) and incubated in pre-heated 37°C PEM buffer [80 mM PIPES, 5 mM EGTA, 2 mM $MgCl_2$, (pH: 6.8)], supplemented with 0.3% Triton-X and 0.125% glutaraldehyde (GA) for 30 s. Next, cells were fixed in a PEM buffer supplemented with 2% paraformaldehyde (PFA) for 15m at 37 °C and washed three times with 1x PBS, 2 min each.

Immunofluorescence

All incubations were performed at room temperature with constant agitation unless otherwise specified. Cells were quenched in 5 mM glycine for 10 min. Blocking was conducted for 1 h in 3% bovine serum albumin (BSA) + 0.01% Triton-X in 1x PBS. Indirect immunofluorescence was applied for microtubules and Golgi apparatus visualization. Specifically, cells were incubated with primary antibodies: Anti-α-Tubulin (Sigma-Aldrich, #T9026, 1:250) and GOLGA/GM130 (Proteintech, #11308-1-AP, 1:500) in staining buffer [1% BSA + 0.01% Triton-X in 1x PBS] overnight at 4°C, following by three washes with PBST [0.01% Triton-X in 1x PBS], 2 min each. Cells were then incubated in staining buffer with secondary antibodies: Donkey Anti-Mouse CF488A (Biotium, #20014-1, 1:500) and Donkey anti-Rabbit Alexa Fluor 647 (Thermo Fisher Scientific, #A-31573, 1:500) for 1 h, and three washes with PBST were repeated. Actin filaments were stained with Phalloidin Conjugate CF568 (Biotium, #00044, 1:50) in 1x PBS for 1h. Finally, cells were incubated in DAPI nuclear dye (Thermo Fisher Scientific, #62248, 300 nM) in 1x PBS for 10 min. Samples were stored at 4°C in 1x PBS with 0.02% sodium azide (NaN_3) until imaging.

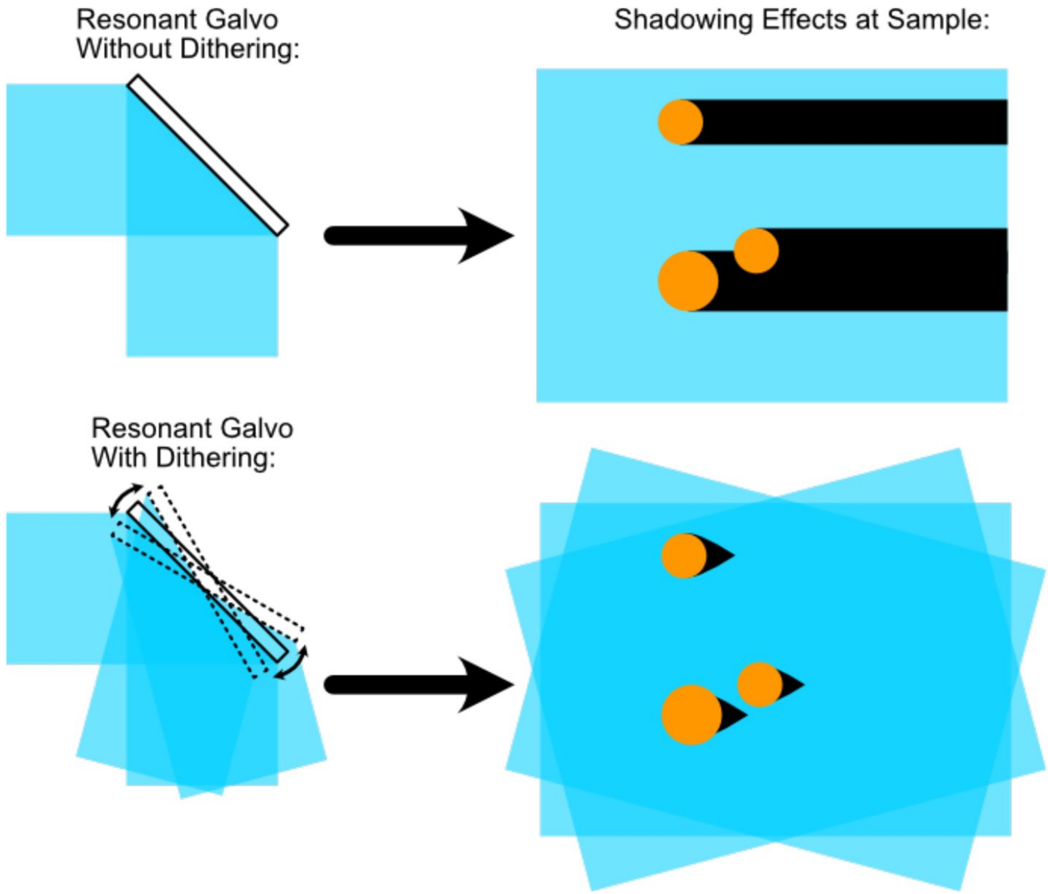
Preparation of Fluorescent Bead Samples

A 5 mm glass coverslip was placed inside a petri dish, and approximately 100 µL of (3-Aminopropyl)triethoxysilane (APTS) at a concentration of 5 mM was applied to its surface. The APTS was incubated for 10–30 min to promote bead adhesion, after which the coverslip was lightly washed three times with deionized water. Fluorescent beads (Fluoresbrite YG Microspheres 0.10 µm, Polyscience, Inc) were diluted to the desired concentration (typically 10^{-3} or 10^{-4} for a normal distribution, 10^{-6} for a sparse distribution) and applied to the treated coverslip, where they were incubated for 2–20 min, with longer incubation times increasing bead density. Finally, the coverslip was lightly washed with DI water to remove unbound beads before imaging.

Custom Machining and Fabrication

All metal components were machined from 6061-T6x aluminum by Protolabs or Xometry, adhering to standard machining tolerances of ±0.005 inches. The sample chamber was 3D printed using a Formlabs Form 3B resin printer with standard black resin. CAD files for all custom parts are provided as a supplementary file, with up-to-date versions available on GitHub.

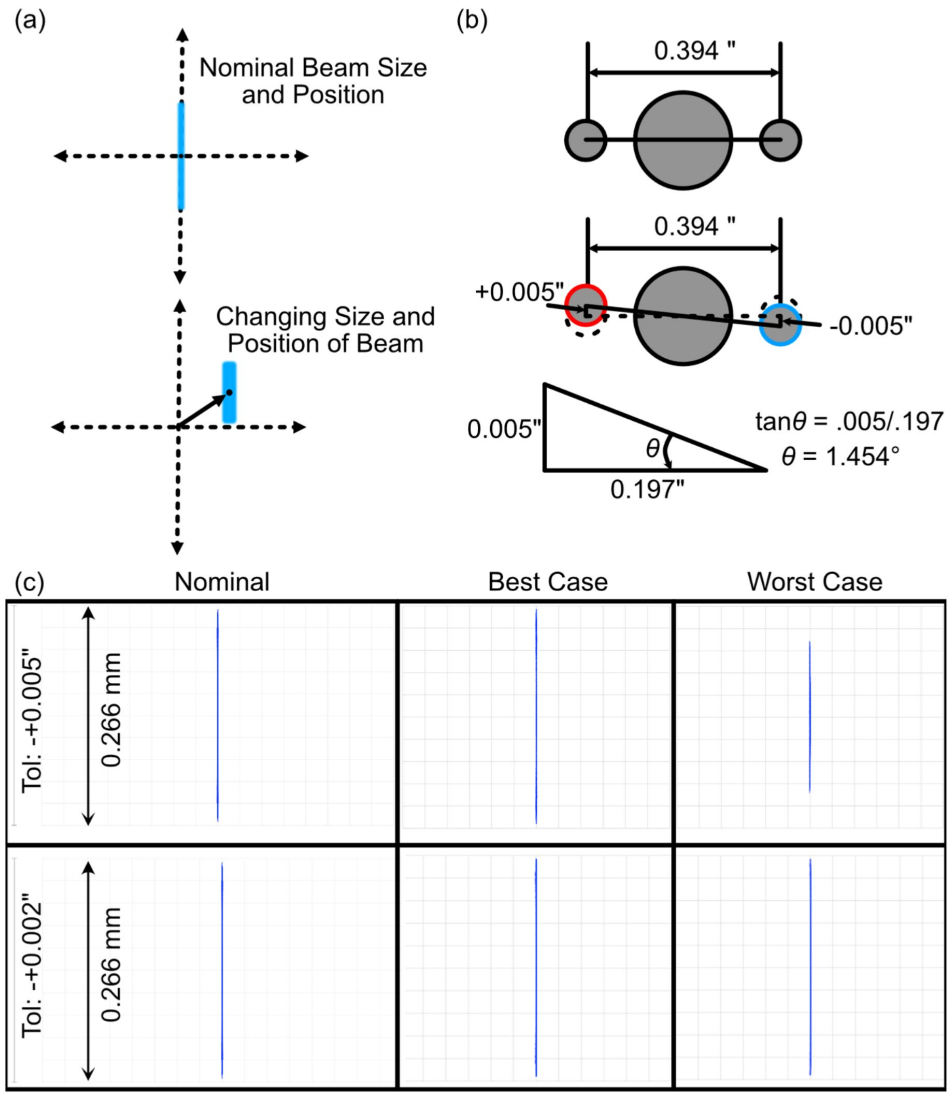
Supplementary figures and tables



Supplementary Figure 1.

Function of the resonant galvo unit, where without dithering, objects in a sample can cast shadows.

Incorporation of dithering at a high speed over the image acquisition time effectively averages the effects of these shadows out of the final image.

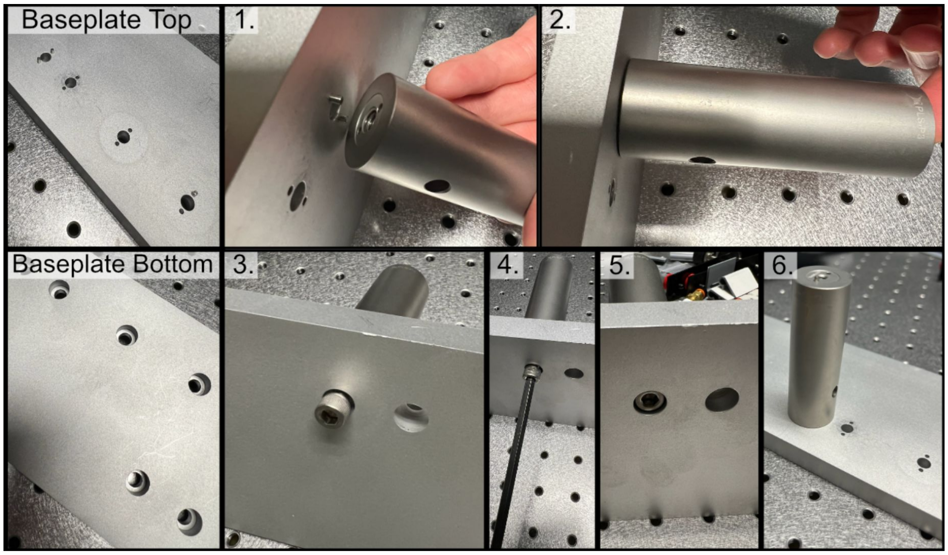


Supplementary Figure 2.

(a) Depiction of the merit function criteria used in our tolerance analysis, where we observed how the beam profile in the perturbed instances changes in both size and position. **(b)** Schematic of the Polaris dowel pin mounting configuration when considering machining tolerances, where in a worst-case scenario the angle offset would be 1.454 degrees. **(c)** Nominal, best case, and worst-case beam profiles in *xz* for both coarse (+-0.005”, top row) and fine (+-0.002”, bottom row) machining tolerances.

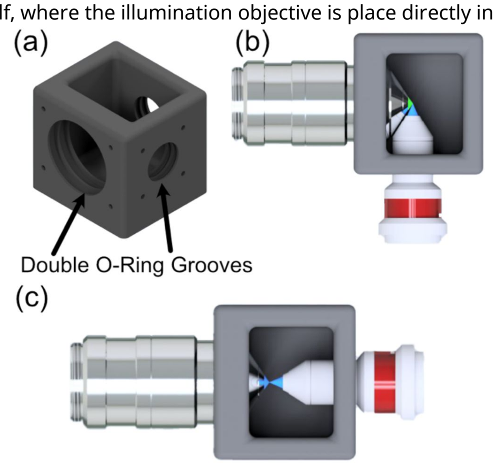
Supplementary Figure 3.

Process of affixing a post to the baseplate, where one first places the post onto dowel pins inserted into the corresponding holes and then fixes the post to the baseplate with a screw.



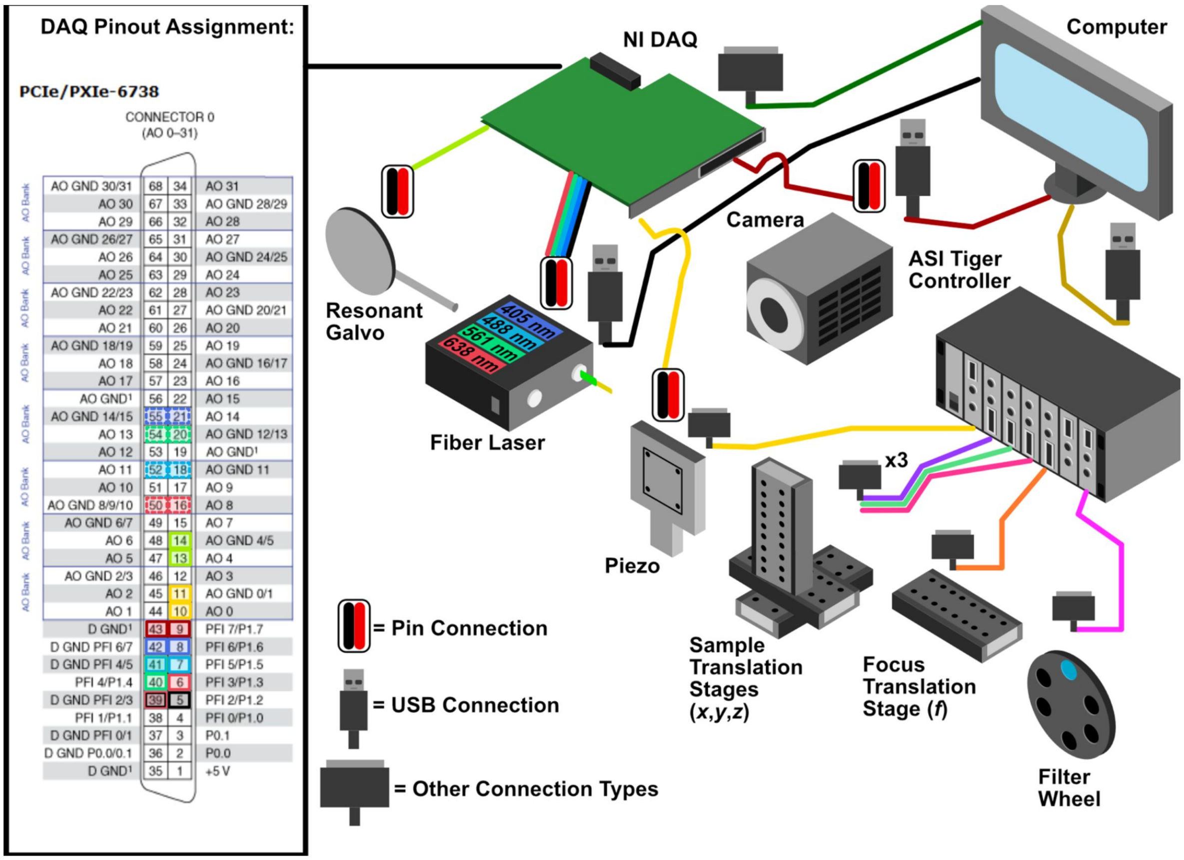
Supplementary Figure 4.

(a) CAD rendering of our custom sample chamber, featuring three possible objective ports, each with two sets of O-rings to ensure a liquid-proof seal. **(b)** Top-down rendering of the traditional imaging configuration for the system, where the illumination and detection objectives are placed orthogonal to one another. **(c)** The second imaging configuration of the system used to image the beam itself, where the illumination objective is place directly in front of the detection objective.



Supplementary Figure 5.

General wiring diagram of the system showing all of the optoelectrical and optomechanical components used, and an inset showing how these components are wired into the NI DAQ.



Supplementary Table S1.

Electrical pinouts used on National Instruments PCIe-6738 data acquisition card.

All analog and digital connections were made using a National Instruments SCB-68A shielded terminal block.

Component	Analog/Digital?	Ground Pin	Active Pin
Resonant Galvo	Analog	14	13
Piezo Motor	Analog	11	10
Camera	Digital	43	9
Fiber Laser (FL) Output Shutter	Digital	39	5
FL λ = 405 nm AO Port	Analog	55	21
FL λ = 405 nm IO Port	Digital	42	8
FL λ = 488 nm AO Port	Analog	18	52
FL λ = 488 nm IO Port	Digital	41	7
FL λ = 561 nm AO Port	Analog	20	54
FL λ = 561 nm IO Port	Digital	41	40
FL λ = 638 nm AO Port	Analog	50	16
FL λ = 638 nm IO Port	Digital	39	6

Name	Manufacturer	Description	Qty	Approximate Cost Per
Shared Equipment				
TG16-BASIC	ASI	Tiger Controller – 16 Bay System	1	$6,350
PCIe-6738	NI	Data Acquisition Card	1	$2,200
SCB-68A	NI	Noise Rejecting Terminal Block	1	$500
SHC68-68-A2	NI	Test Cable Assembly	1	$250
784-736-02R	TMC	36x60x18” Performance Series Optical Top	1	$6,275
14UD-42X-24	TMC	Ultradamp Vibration Isolation with Casters	1	$9,300
SX6300	Colfax International	Colfax SX6300 Workstation	1	$6,800
Detection Path				
LS-100-AMCCH	ASI	100 mm Linear Focusing Stage, 16 TPI	1	$2,250
TGDCM2	ASI	2-Axis Stage Control Card	1	$1,050
C60-EXT-15	ASI	15 mm Tube Extension	1	$75
RAO-0051	ASI	M32x0.75 Threaded Sleeve	1	$150
FW-0002-8	ASI	8-Position Filter Wheel	1	$3,500
FW-C-MNT-K1	ASI	Filter Wheel to MIM Adapter Kit	1	$400
C60-TUBE-400	ASI	400 mm Achromatic Tube Lens	1	$875
TGFW	ASI	Filter Wheel Control Card	1	$1,150
C13440-20CU	Hamamatsu	ORCA Flash4.0 V3	1	$16,500
OH-CAMRA-1007	Hamamatsu	Firebird CamLink Board	1	$2,000
N25X-APO-MP	Thorlabs	Nikon 25x NA 1.1 Detection Objective	1	$33,810
445/20-25nm	Semrock	Brightline Bandpass filter	1	$375

Supplementary Table S2.

Detailed equipment list.

Prices are approximate and subject to change-often unpredictably due to economically self-defeating trade wars. *Indicates that the part was custom ordered from Thorlabs.

529/24nm	Semrock	Brightline Bandpass filter	2	$375
605/15-25nm	Semrock	Brightline Bandpass filter	1	$375
676/29-25nm	Semrock	Brightline Bandpass filter	1	$375
Sample Positioning				
LS-5012	ASI	Breadboard Adapter	1	$160
LS-5013	ASI	Right Angle Bracket	1	$250
DV-6010-C	ASI	Dovetail Mount Pair	1	$230
LS-50-AMCLLS	ASI	50 mm Linear Stage with Stainless Steel Slide	1	$2,425
LS-100-AMERL	ASI	100 mm Linear Stage, 16 TPI, Extended, Right	1	$3,100
LS-50-AMELL	ASI	50 mm Linear Stage	1	$2,250
TGADEPT	ASI	Piezo Control Card	1	$1,500
HS1.100	PiezoConcept	Piezo Concept 100 Micron Piezo Stage	1	$4,100
ADAPTHS1BB	PiezoConcept	Adapter Plate for PiezoConcept HS1	1	$120
Angle Bracket Adapter	Xometry	Angle adapter for mounting piezo at an angle	1	$592.58
Illumination Path				
L4CC	Oxxius	Multicolor Laser Source	1	25,000
P3-405B-FC-1	Thorlabs	Fiber Cable Single Mode FC/APC	1	$120
CFC11A-A	Thorlabs	Fiber Collimator	1	$370
Polaris-K1XY	Thorlabs	Kinematic Mount for Fiber Laser Collimator	1	$1375
AC254-030-A-ML	Thorlabs	Achromatic Lens f=30mm (L1)	1	$125
AC254-080-A-ML	Thorlabs	Achromatic Lens f=80mm (L2)	1	$115
ACY254-075-A	Thorlabs	Achromatic Cylindrical Lens f=75mm (L3)	1	$445
AC254-250-A-ML	Thorlabs	Achromatic Lens f=250mm (L4)	1	$90
PF10-03-P01	Thorlabs	1” Protected Silver Mirror	1	$55
TL20X-MPL	Thorlabs	Illumination Objective	1	$4820
VA100CP	Thorlabs	Rectangular Aperture	1	$315
IDA12	Thorlabs	Circular Aperture Iris	1	$60
Polaris-P150	Thorlabs	1” Diameter Polaris Mounting Post, 1.5” Length	1	$45
Polaris-P2	Thorlabs	1” Diameter Polaris Mounting Post, 2” Length	1	$45
Polaris-P3	Thorlabs	1” Diameter Polaris Mounting Posts, 3” Length	3	$45
Polaris-P075*	Thorlabs	1” Diameter Polaris Mounting Post, 0.75” Length	1	$45
Polaris-P225*	Thorlabs	1” Diameter Polaris Mounting Post, 2.25” Length	1	$45
Polaris-P250*	Thorlabs	1” Diameter Polaris Mounting Post, 2.5” Length	2	$45
Polaris-MA45	Thorlabs	Polaris 45 Degree Adapter for Mirror Mount	1	$50
Polaris-B1S	Thorlabs	Polaris Flexure-Closed Lens Mount	3	$110
Polaris-1XY	Thorlabs	Polaris XY Translation Mount for Illumination Objective	1	$1020
Polaris-K1S4	Thorlabs	Polaris 1” Mirror Mount	1	$185

Supplementary Table S2. (continued)

Supplementary Table S2. (continued)

SM1A12	Thorlabs	Thread Adapter for Illumination Objective into Polaris-1XY	1	$25
6SC04KA040-01Y	Novanta	1-Axis 4 kHz Resonant Galvo and Servo	1	$3,700
Galvo Holder	Protolabs	Holder for Galvo	1	$400
TGPOW-12-3	ASI	Galvo Low Noise Power Supply +12V 3A	1	$950
RSP1 Adapter	Xometry	Custom Adapter for Thorlabs RSP1 to Polaris Posts	1	$80
VA100CP Adapter	Xometry	Custom Adapter for Thorlabs VA100CP to Polaris Posts	1	$80
CP02 Adapter	Xometry	Custom Adapter for Thorlabs CP02 to Polaris Posts	1	$80
Illumination Path Baseplate	Xometry	Baseplate for Illumination Path	1	$1000

Supplementary Table S3.

Approximate Cost.

Group	Approximate Cost
Illumination Path	$41,195
Sample Positioning	$14,727
Detection Path	$63,635
Shared Equipment	$31,675
Total:	$151,232

Acknowledgements

The authors would like to thank Calvin Jones and Dr. Sophia Theodossiou (Boise State University) for their assistance in designing and printing the custom sample chamber, and Melissa Glidewell for her initial evaluation of optical tolerances. This work was supported by the National Institutes of Health (U54 CA268072 and RM1 GM145399).

Additional information

Author Contributions

J.H., formal analysis, investigation, methodology, validation, visualization, and writing. S.G., resources, visualization, and writing. K.M.D. funding acquisition, investigation, methodology, project administration, software, resources, supervision, visualization, and writing.

Additional files

Supplementary Documentation.

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Reviewer #1 (Public review):

Summary:

The article presents the details of the high-resolution light-sheet microscopy system developed by the group. In addition to presenting the technical details of the system, its resolution has been characterized and its functionality demonstrated by visualizing subcellular structures in a biological sample.

Strengths:

(1) The article includes extensive supplementary material that complements the information in the main article.

(2) However, in some sections, the information provided is somewhat superficial.

Weaknesses:

(1) Although a comparison is made with other light-sheet microscopy systems, the presented system does not represent a significant advance over existing systems. It uses high numerical aperture objectives and Gaussian beams, achieving resolution close to theoretical after deconvolution. The main advantage of the presented system is its ease of construction, thanks to the design of a perforated base plate.

(2) Using similar objectives (Nikon 25x and Thorlabs 20x), the results obtained are similar to those of the LLSM system (using a Gaussian beam without laser modulation). However, the article does not mention the difficulties of mounting the sample in the implemented configuration.

(3) The authors present a low-cost, open-source system. Although they provide open source code for the software (navigate), the use of proprietary electronics (ASI, NI, etc.) makes the system relatively expensive. Its low cost is not justified.

(4) The fibroblast images provided are of exceptional quality. However, these are fixed samples. The system lacks the necessary elements for monitoring cells in vivo, such as temperature or pH control.

https://doi.org/10.7554/eLife.106910.1.sa3

Reviewer #2 (Public review):

Summary:

The authors present Altair-LSFM (Light Sheet Fluorescence Microscope), a high-resolution, open-source microscope, that is relatively easy to align and construct and achieves sub-cellular resolution. The authors developed this microscope to fill a perceived need that current open-source systems are primarily designed for large specimens and lack sub-cellular resolution or are difficult to construct and align, and are not stable. While commercial alternatives exist that offer sub-cellular resolution, they are expensive. The authors' manuscript centers around comparisons to the highly successful lattice light-sheet microscope, including the choice of detection and excitation objectives. The authors thus claim that there remains a critical need for high-resolution, economical, and easy-to-implement LSFM systems.

Strengths:

The authors succeed in their goals of implementing a relatively low-cost (~ USD 150K) open-source microscope that is easy to align. The ease of alignment rests on using custom-designed baseplates with dowel pins for precise positioning of optics based on computer analysis of opto-mechanical tolerances, as well as the optical path design. They simplify the excitation optics over Lattice light-sheet microscopes by using a Gaussian beam for illumination while maintaining lateral and axial resolutions of 235 and 350 nm across a 260-um field of view after deconvolution. In doing so they rest on foundational principles of optical microscopy that what matters for lateral resolution is the numerical aperture of the detection objective and proper sampling of the image field on to the detection, and the axial resolution depends on the thickness of the light-sheet when it is thinner than the depth of field of the detection objective. This concept has unfortunately not been completely clear to users of high-resolution light-sheet microscopes and is thus a valuable demonstration. The microscope is controlled by an open-source software, Navigate, developed by the authors, and it is thus foreseeable that different versions of this system could be implemented depending on experimental needs while maintaining easy alignment and low cost. They demonstrate system performance successfully by characterizing their sheet, point-spread function, and visualization of sub-cellular structures in mammalian cells, including microtubules, actin filaments, nuclei, and the Golgi apparatus.

Weaknesses:

There is a fixation on comparison to the first-generation lattice light-sheet microscope, which has evolved significantly since then:

(1) The authors claim that commercial lattice light-sheet microscopes (LLSM) are "complex, expensive, and alignment intensive", I believe this sentence applies to the open-source version of LLSM, which was made available for wide dissemination. Since then, a commercial solution has been provided by 3i, which is now being used in multiple cores and labs but does require routine alignments. However, Zeiss has also released a commercial turn-key system, which, while expensive, is stable, and the complexity does not interfere with the experience of the user. Though in general, statements on ease of use and stability might be considered anecdotal and may not belong in a scientific article, unreferenced or without data.

(2) One of the major limitations of the first generation LLSM was the use of a 5 mm coverslip, which was a hinderance for many users. However, the Zeiss system elegantly solves this problem, and so does Oblique Plane Microscopy (OPM), while the Altair-LSFM retains this feature, which may dissuade widespread adoption. This limitation and how it may be overcome in future iterations is not discussed.

(3) Further, on the point of sample flexibility, all generations of the LLSM, and by the nature of its design, the OPM, can accommodate live-cell imaging with temperature, gas, and humidity control. It is unclear how this would be implemented with the current sample chamber. This limitation would severely limit use cases for cell biologists, for which this microscope is designed. There is no discussion on this limitation or how it may be overcome in future iterations.

(4) The authors' comparison to LLSM is constrained to the "square" lattice, which, as they point out, is the most used optical lattice (though this also might be considered anecdotal). The LLSM original design, however, goes far beyond the square lattice, including hexagonal lattices, the ability to do structured illumination, and greater flexibility in general in terms of light-sheet tuning for different experimental needs, as well as not being limited to just sample scanning. Thus, the Alstair-LSFM cannot compare to the original LLSM in terms of versatility, even if comparisons to the resolution provided by the square lattice are fair.

(5) There is no demonstration of the system's live-imaging capabilities or temporal resolution, which is the main advantage of existing light-sheet systems.

While the microscope is well designed and completely open source, it will require experience with optics, electronics, and microscopy to implement and align properly. Experience with custom machining or soliciting a machine shop is also necessary. Thus, in my opinion, it is unlikely to be implemented by a lab that has zero prior experience with custom optics or can hire someone who does. Altair-LSFM may not be as easily adaptable or implementable as the authors describe or perceive in any lab that is interested, even if they can afford it. The authors indicate they will offer "workshops," but this does not necessarily remove the barrier to entry or lower it, perhaps as significantly as the authors describe.

There is a claim that this design is easily adaptable. However, the requirement of custom-machined baseplates and in silico optimization of the optical path basically means that each new instrument is a new design, even if the Navigate software can be used. It is unclear how

Altair-LSFM demonstrates a modular design that reduces times from conception to optimization compared to previous implementations.

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Reviewer #3 (Public review):

Summary:

This manuscript introduces a high-resolution, open-source light-sheet fluorescence microscope optimized for sub-cellular imaging.

The system is designed for ease of assembly and use, incorporating a custom-machined baseplate and in silico optimized optical paths to ensure robust alignment and performance. The authors demonstrate lateral and axial resolutions of ~235 nm and ~350 nm after deconvolution, enabling imaging of sub-diffraction structures in mammalian cells.

The important feature of the microscope is the clever and elegant adaptation of simple gaussian beams, smart beam shaping, galvo pivoting and high NA objectives to ensure a uniform thin light-sheet of around 400 nm in thickness, over a 266 micron wide Field of view, pushing the axial resolution of the system beyond the regular diffraction limited-based tradeoffs of light-sheet fluorescence microscopy.

Compelling validation using fluorescent beads and multicolor cellular imaging highlights the system's performance and accessibility. Moreover, a very extensive and comprehensive manual of operation is provided in the form of supplementary materials. This provides a DIY blueprint for researchers who want to implement such a system.

Strengths:

(1) Strong and accessible technical innovation:

With an elegant combination of beam shaping and optical modelling, the authors provide a high-resolution light-sheet system that overcomes the classical light-sheet tradeoff limit of a thin light-sheet and a small field of view. In addition, the integration of in silico modelling with a custom-machined baseplate is very practical and allows for ease of alignment procedures. Combining these features with the solid and super-extensive guide provided in the supplementary information, this provides a protocol for replicating the microscope in any other lab.

(2) Impeccable optical performance and ease of mounting of samples:

The system takes advantage of the same sample-holding method seen already in other implementations, but reduces the optical complexity. At the same time, the authors claim to achieve similar lateral and axial resolution to Lattice-light-sheet microscopy (although without a direct comparison (see below in the "weaknesses" section). The optical characterization of the system is comprehensive and well-detailed. Additionally, the authors validate the system imaging sub-cellular structures in mammalian cells.

(3) Transparency and comprehensiveness of documentation and resources:

A very detailed protocol provides detailed documentation about the setup, the optical modeling, and the total cost.

Weaknesses:

(1) Limited quantitative comparisons:

Although some qualitative comparison with previously published systems (diSPIM, lattice light-sheet) is provided throughout the manuscript, some side-by-side comparison would be of great benefit for the manuscript, even in the form of a theoretical simulation. While having a direct imaging comparison would be ideal, it's understandable that this goes beyond the interest of the paper; however, a table referencing image quality parameters (taken from the literature), such as signal-to-noise ratio, light-sheet thickness, and resolutions, would really enhance the features of the setup presented. Moreover, based also on the necessity for optical simplification, an additional comment on the importance/difference of dual objective/single objective light-sheet systems could really benefit the discussion.

(2) Limitation to a fixed sample:

In the manuscript, there is no mention of incubation temperature, CO_2 regulation, Humidity control, or possible integration of commercial environmental control systems. This is a major limitation for an imaging technique that owes its popularity to fast, volumetric, live-cell imaging of biological samples.

(3) System cost and data storage cost:

While the system presented has the advantage of being open-source, it remains relatively expensive (considering the 150k without laser source and optical table, for example). The manuscript could benefit from a more direct comparison of the performance/cost ratio of existing systems, considering academic settings with budgets that most of the time would not allow for expensive architectures. Moreover, it would also be beneficial to discuss the adaptability of the system, in case a 30k objective could not be feasible. Will this system work with different optics (with the obvious limitations coming with the lower NA objective)? This could be an interesting point of discussion. Adaptability of the system in case of lower budgets or more cost-effective choices, depending on the needs.

Last, not much is said about the need for data storage. Light-sheet microscopy's bottleneck is the creation of increasingly large datasets, and it could be beneficial to discuss more about the storage needs and the quantity of data generated.

Conclusion:

Altair-LSFM represents a well-engineered and accessible light-sheet system that addresses a longstanding need for high-resolution, reproducible, and affordable sub-cellular light-sheet imaging. While some aspects-comparative benchmarking and validation, limitation for fixed samples-would benefit from further development, the manuscript makes a compelling case for Altair-LSFM as a valuable contribution to the open microscopy scientific community.

https://doi.org/10.7554/eLife.106910.1.sa1

Author response:

> ***eLife Assessment***
>
> *This useful study presents Altair-LSFM, a solid and well-documented implementation of a light-sheet fluorescence microscope (LSFM) designed for accessibility and cost reduction. While the approach offers strengths such as the use of custom-machined baseplates and detailed assembly instructions, its overall impact is limited by the lack of live-cell imaging capabilities and the absence of a clear, quantitative comparison to existing LSFM platforms. As such, although technically competent, the broader utility and uptake of this system by the community may be limited.*

We thank the reviewers and editors for their thoughtful evaluation of our work and for recognizing the technical strengths of the Altair-LSFM platform, including the custom-machined baseplates and detailed documentation provided to support accessibility and reproducibility. We respectfully disagree, however, with the assessment that the system lacks live-cell imaging capabilities. We are fully confident in the system's suitability for live-cell applications and will demonstrate this by including representative live-cell imaging data in the revised manuscript, along with detailed instructions for implementing environment control. Moreover, we will expand our discussion to include a broader, more quantitative comparison to existing LSFM platforms—highlighting trade-offs in cost, performance, and accessibility—to better contextualize Altair's utility and adaptability across diverse research settings.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The article presents the details of the high-resolution light-sheet microscopy system developed by the group. In addition to presenting the technical details of the system, its resolution has been characterized and its functionality demonstrated by visualizing subcellular structures in a biological sample.*
>
> *Strengths:*
>
> *(1) The article includes extensive supplementary material that complements the information in the main article.*
>
> *(2) However, in some sections, the information provided is somewhat superficial.*

Our goal was to make the supplemental content as comprehensive and useful as possible. In addition to the materials provided with the manuscript, our intention is for the online documentation (available at thedeanlab.github.io/altair) to serve as a living resource that evolves in response to user feedback. For this reason, we are especially interested in identifying and expanding any sections that are perceived as superficial, and we would greatly appreciate the reviewer's guidance on which areas would benefit from further elaboration.

> *Weaknesses:*
>
> *(1) Although a comparison is made with other light-sheet microscopy systems, the presented system does not represent a significant advance over existing systems. It uses high numerical aperture objectives and Gaussian beams, achieving resolution close to theoretical after deconvolution. The main advantage of the presented system is its ease of construction, thanks to the design of a perforated base plate.*

We appreciate the reviewer's assessment and the opportunity to clarify our intent. Our primary goal was not to introduce new optical functionality beyond that of existing high-performance light-sheet systems, but rather to reduce the barrier to entry for non-specialist labs.

> *(2) Using similar objectives (Nikon 25x and Thorlabs 20x), the results obtained are similar to those of the LLSM system (using a Gaussian beam without laser modulation). However, the article does not mention the difficulties of mounting the sample in the implemented configuration.*

We agree that there are practical challenges associated with handling 5 mm diameter coverslips. However, the Nikon 25x can readily be replaced by a Zeiss W Plan-Apochromat 20x/1.0 objective, which eliminates the need for the 5 mm coverslip[1]. In the revised manuscript, we will more explicitly detail the practical challenges in handling a 5 mm coverslip and mention the alternative detection objective.

> *(3) The authors present a low-cost, open-source system. Although they provide open source code for the software (navigate), the use of proprietary electronics (ASI, NI, etc.) makes the system relatively expensive. Its low cost is not justified.*

We understand the reviewer's concern regarding the use of proprietary control hardware such as the ASI Tiger Controller and NI data acquisition cards. While lower-cost alternatives for analog and digital control (e.g., microcontroller-based systems) do exist, our choice was intentional. By relying on a unified and professionally supported platform, we minimize the complexity of sourcing, configuring, and integrating components from disparate vendors—each of which would otherwise demand specialized technical expertise. Moreover, in future releases, we aim to further streamline the system by eliminating the need for the NI card, consolidating all optoelectronic control through the ASI Tiger Controller. This approach allows users to purchase a fully assembled and pre-configured system that can be operational with minimal effort.

It is worth noting that the ASI components are not the primary cost driver. The full set—including XYZ and focusing stages, a filter wheel, a tube lens, the Tiger Controller, and basic optomechanical adapters—costs approximately $27,000, or ~18% of the total system cost. Additional cost reductions are possible. For example, replacing the motorized sample positioning and focusing stages with manual alternatives could reduce the cost by ~$12,000. However, this would eliminate key functionality such as autofocusing, 3D tiling, and multi-position acquisition. Open-source mechanical platforms such as OpenFlexure could in principle be adapted, but they would require custom assembly and would need to be integrated into our control software. Similarly, the filter wheel could be omitted in favor of a multi-band emission filter, reducing the cost by ~$5,000. However, this comes at the expense of increased spectral crosstalk, often necessitating spectral unmixing. An industrial CMOS camera—such as the Ximea MU196CR-ON, recently demonstrated in a Direct View Oblique Plane Microscopy configuration[2]—could substitute for the sCMOS cameras typically used in high-end imaging. However, these industrial sensors often exhibit higher noise floors and lower dynamic range, limiting sensitivity for low-signal imaging applications.

While a $150,000 system represents a significant investment, we consider it relatively cost-effective in the context of advanced light-sheet microscopy. For comparison, commercially available systems with similar optical performance—such as LLSM systems from 3i or Zeiss—are several-fold more expensive.

> *(4) The fibroblast images provided are of exceptional quality. However, these are fixed samples. The system lacks the necessary elements for monitoring cells in vivo, such as temperature or pH control.*

We thank the reviewer for their positive comment regarding the quality of our fibroblast images. As noted, the current manuscript focuses on the optical design and performance characterization of the system, using fixed specimens to validate resolution and imaging stability. We acknowledge the importance of environmental control for live-cell imaging. Temperature regulation is routinely implemented in our lab using flexible adhesive heating elements paired with a power supply and PID controller. For pH stabilization in systems that lack a 5% CO_2 atmosphere, we typically supplement the imaging medium with 10–25 mM HEPES buffer. In the revised manuscript, we will introduce a modified sample chamber capable of maintaining user-specified temperatures, along with detailed assembly instructions. We will also include representative live-cell imaging data to demonstrate the feasibility of in vitro imaging using this system.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The authors present Altair-LSFM (Light Sheet Fluorescence Microscope), a high-resolution, open-source microscope, that is relatively easy to align and construct and achieves sub-cellular resolution. The authors developed this microscope to fill a perceived need that current open-source systems are primarily designed for large specimens and lack sub-cellular resolution or are difficult to construct and align, and are not stable. While commercial alternatives exist that offer sub-cellular resolution, they are expensive. The authors' manuscript centers around comparisons to the highly successful lattice light-sheet microscope, including the choice of detection and excitation objectives. The authors thus claim that there remains a critical need for high-resolution, economical, and easy-to-implement LSFM systems.*
>
> *Strengths:*
>
> *The authors succeed in their goals of implementing a relatively low-cost (~ USD 150K) open-source microscope that is easy to align. The ease of alignment rests on using custom-designed baseplates with dowel pins for precise positioning of optics based on computer analysis of opto-mechanical tolerances, as well as the optical path design. They simplify the excitation optics over Lattice light-sheet microscopes by using a Gaussian beam for illumination while maintaining lateral and axial resolutions of 235 and 350 nm across a 260-um field of view after deconvolution. In doing so they rest on foundational principles of optical microscopy that what matters for lateral resolution is the numerical aperture of the detection objective and proper sampling of the image field on to the detection, and the axial resolution depends on the thickness of the light-sheet when it is thinner than the depth of field of the detection objective. This concept has unfortunately not been completely clear to users of high-resolution light-sheet microscopes and is thus a valuable demonstration. The microscope is controlled by an open-source software, Navigate, developed by the authors, and it is thus foreseeable that different versions of this system could be implemented depending on experimental needs while maintaining easy alignment and low cost. They demonstrate system performance successfully by characterizing their sheet, point-spread function, and visualization of sub-cellular structures in mammalian cells, including microtubules, actin filaments, nuclei, and the Golgi apparatus.*

We thank the reviewer for their thoughtful summary of our work. We are pleased that the foundational optical principles, design rationale, and emphasis on accessibility came through clearly. We agree that the approach used to construct the microscope is highly modular, and we anticipate that these design principles will serve as the basis for additional system variants tailored to specific biological samples and experimental contexts. To support this, we provide all Zemax simulations and CAD files openly on our GitHub repository, enabling advanced users to build upon our design and create new functional variants of the Altair system.

> *Weaknesses:*
>
> *There is a fixation on comparison to the first-generation lattice light-sheet microscope, which has evolved significantly since then:*

> *(1) The authors claim that commercial lattice light-sheet microscopes (LLSM) are "complex, expensive, and alignment intensive", I believe this sentence applies to the open-source version of LLSM, which was made available for wide dissemination. Since then, a commercial solution has been provided by 3i, which is now being used in multiple cores and labs but does require routine alignments. However, Zeiss has also released a commercial turn-key system, which, while expensive, is stable, and the complexity does not interfere with the experience of the user. Though in general, statements on ease of use and stability might be considered anecdotal and may not belong in a scientific article, unreferenced or without data.*

The referee is correct that our comparisons reference the original LLSM design, which was simultaneously disseminated as an open-source platform and commercialized by 3i. While we acknowledge that newer variants of LLSM have been developed—including systems incorporating adaptive optics[3] and the MOSAIC platform (which remains unpublished)—the original implementation remains the most widely described and cited in the literature. It is therefore the most appropriate point of comparison for contextualizing Altair's performance, complexity, and accessibility. Importantly, this version of LLSM is far from obsolete; it continues to be one of the most commonly used imaging systems at Janelia Research Campus's Advanced Imaging Center.

We acknowledge that more recent commercial implementation by Zeiss has addressed several of the practical limitations associated with the original design. In particular, we agree that the Zeiss Lattice Lightsheet 7 system, which integrates a meniscus lens to facilitate oblique imaging through a coverslip, offers a user-friendly experience—albeit with a modest tradeoff in resolution (reported deskewed resolution: 330 nm × 330 nm × 500–1000 nm).

While we recognize that statements on usability and stability can be subjective, one objective proxy for system complexity is the number of optical elements that require precise alignment during assembly. The original LLSM setup includes approximately 29 optical components that must each be carefully positioned laterally, angularly, and coaxially along the optical path. In contrast, the first-generation Altair system contains only 9 such elements. By this metric, Altair is considerably simpler to assemble and align, supporting our overarching goal of making high-resolution light-sheet imaging more accessible to non-specialist laboratories. In the revised manuscript, we will clarify the scope of our comparison and provide more precise language about what we mean by complexity (e.g., number of optical elements needed to align).

> *(2) One of the major limitations of the first generation LLSM was the use of a 5 mm coverslip, which was a hinderance for many users. However, the Zeiss system elegantly solves this problem, and so does Oblique Plane Microscopy (OPM), while the Altair-LSFM retains this feature, which may dissuade widespread adoption. This limitation and how it may be overcome in future iterations is not discussed.*

We agree that the use of 5 mm diameter coverslips, while enabling high-NA imaging in the current Altair-LSFM configuration, may serve as an inconvenience for many users. We will discuss this more explicitly in the revised manuscript. Specifically, we note that changing the detection objective is sufficient to eliminate the need for a 5 mm coverslip. For example, as demonstrated in Moore et al., Lab Chip 2021, pairing the Zeiss W Plan-Apochromat 20x/1.0 objective with the Thorlabs TL20X-MPL allows imaging beyond the physical surfaces of both objectives, removing the constraint imposed by small-format coverslips[1]. In the revised manuscript, we will propose this modification as a straightforward path for increasing compatibility with more conventional sample mounting formats.

> *(3) Further, on the point of sample flexibility, all generations of the LLSM, and by the nature of its design, the OPM, can accommodate live-cell imaging with temperature, gas, and humidity control. It is unclear how this would be implemented with the current sample chamber. This limitation would severely limit use cases for cell biologists, for which this microscope is designed. There is no discussion on this limitation or how it may be overcome in future iterations.*

We appreciate the reviewer's emphasis on the importance of environmental control for live-cell imaging applications. It is worth noting that the original LLSM design, including the system commercialized by 3i, provided temperature control only, without integrated gas or humidity regulation. Despite this, it has been successfully used by a wide range of scientists to generate important biological insights.

We agree that both OPM and the Zeiss implementation of LLSM offer clear advantages in terms of environmental control, as we previously discussed in detail in Sapoznik et al., eLife, 2020[4]. However, assembly of high numerical aperture OPM systems is highly technical, and no open-source variant of OPM delivers sub-cellular scale resolution yet.

> *(4) The authors' comparison to LLSM is constrained to the "square" lattice, which, as they point out, is the most used optical lattice (though this also might be considered anecdotal). The LLSM original design, however, goes far beyond the square lattice, including hexagonal lattices, the ability to do structured illumination, and greater flexibility in general in terms of light-sheet tuning for different experimental needs, as well as not being limited to just sample scanning. Thus, the Alstair-LSFM cannot compare to the original LLSM in terms of versatility, even if comparisons to the resolution provided by the square lattice are fair.*

We thank the reviewer for this comment. It is true that our discussion focused primarily on the square lattice implementation of LLSM. While this could be viewed as a subset of the system's broader capabilities, we chose this focus intentionally, as the square lattice remains by far the most commonly used variant in practice. Even in the original LLSM publication, 16 out of 20 figure subpanels utilized the square lattice, with only one panel each representing the hexagonal lattice in SIM mode, a standard Bessel beam in incoherent SIM mode, a hex lattice in dithered mode, and a single Bessel in dithered mode. This usage pattern largely reflects the operational simplicity of the square lattice: it minimizes sidelobe growth and enables more straightforward alignment and data processing compared to hexagonal or structured illumination modes.

In 2019, we performed an exhaustive accounting of published illumination modes in LLSM and found that the SIM mode had only been used in two additional peer-reviewed publications at that time. We will consider updating this table in the revised manuscript and will expand our discussion to acknowledge the broader flexibility of the LLSM platform—including its capacity for structured illumination and alternative light-sheet geometries. However, we will also emphasize that, despite these advanced capabilities, the square lattice remains the dominant mode used by the community and therefore serves as a fair and practical benchmark for comparison.

> *(5) There is no demonstration of the system's live-imaging capabilities or temporal resolution, which is the main advantage of existing light-sheet systems.*

In the revised manuscript, we will include a demonstration of live-cell imaging to directly validate the system's suitability for dynamic biological applications. We will also characterize the temporal resolution of the system. As a sample-scanning microscope, the imaging speed is primarily limited by the performance of the Z-piezo stage. For simplicity and reduced optoelectronic complexity, we currently power the piezo through the ASI Tiger Controller. We will expand the supplementary material to describe the design criteria behind this choice, including potential trade-offs, and provide data quantifying the achievable volume rates under typical operating conditions.

> *While the microscope is well designed and completely open source, it will require experience with optics, electronics, and microscopy to implement and align properly. Experience with custom machining or soliciting a machine shop is also necessary. Thus, in my opinion, it is unlikely to be implemented by a lab that has zero prior experience with custom optics or can hire someone who does. Altair-LSFM may not be as easily adaptable or implementable as the authors describe or perceive in any lab that is interested, even if they can afford it. The authors indicate they will offer "workshops," but this does not necessarily remove the barrier to entry or lower it, perhaps as significantly as the authors describe.*

We appreciate the reviewer's perspective and agree that building any high-performance custom microscope—Altair-LSFM included—requires a baseline familiarity with optics and instrumentation. Our goal is not to eliminate this requirement entirely, but to significantly reduce the technical and logistical barriers that typically accompany custom light-sheet microscope construction.

Importantly, no machining experience or in-house fabrication capabilities are required—users can simply submit provided design files and specifications directly to the vendor. We will make this process as straightforward as possible by supplying detailed instructions, recommended materials, and vendor-ready files. Additionally, we draw encouragement from the success of related efforts such as mesoSPIM, which has seen over 30 successful implementations worldwide using a similar model of exhaustive online documentation, open-source control software, and community support through user meetings and workshops.

We recognize that documentation alone is not always sufficient, and we are committed to further lowering barriers to adoption. To this end, we are actively working with commercial vendors to streamline procurement and reduce the logistical burden on end users. Additionally, Altair-LSFM is supported by a Biomedical Technology Development and Dissemination (BTDD) grant, which provides dedicated resources for hosting workshops, offering real-time community support, and generating supplementary materials such as narrated video tutorials. We will expand our discussion in the revised manuscript to better acknowledge these implementation challenges and outline our ongoing strategies for supporting a broad and diverse user base.

> *There is a claim that this design is easily adaptable. However, the requirement of custom-machined baseplates and in silico optimization of the optical path basically means that each new instrument is a new design, even if the Navigate software can be used. It is unclear how Altair-LSFM demonstrates a modular design that reduces times from conception to optimization compared to previous implementations.*

We appreciate the reviewer's comment and agree that our language regarding adaptability may have been too strong. It was not our intention to suggest that the system can be easily modified without prior experience. Meaningful adaptations of the optical or mechanical design would require users to have expertise in optical layout, optomechanical design, and alignment.

That said, for labs with sufficient expertise, we aim to facilitate such modifications by providing comprehensive resources—including detailed Zemax simulations, CAD models, and alignment documentation. These materials are intended to reduce the development burden for those seeking to customize the platform for specific experimental needs.

In the revised manuscript, we will clarify this point and explicitly state in the discussion what technical expertise is required to modify the system. We will also revise our language around adaptability to better reflect the intended audience and realistic scope of customization.

Reviewer #3 (Public review):

Summary:

This manuscript introduces a high-resolution, open-source light-sheet fluorescence microscope optimized for sub-cellular imaging.

The system is designed for ease of assembly and use, incorporating a custom-machined baseplate and in silico optimized optical paths to ensure robust alignment and performance. The authors demonstrate lateral and axial resolutions of ~235 nm and ~350 nm after deconvolution, enabling imaging of sub-diffraction structures in mammalian cells.

The important feature of the microscope is the clever and elegant adaptation of simple gaussian beams, smart beam shaping, galvo pivoting and high NA objectives to ensure a uniform thin light-sheet of around 400 nm in thickness, over a 266 micron wide Field of view, pushing the axial resolution of the system beyond the regular diffraction limited-based tradeoffs of light-sheet fluorescence microscopy.

Compelling validation using fluorescent beads and multicolor cellular imaging highlights the system's performance and accessibility. Moreover, a very extensive and comprehensive manual of operation is provided in the form of supplementary materials. This provides a DIY blueprint for researchers who want to implement such a system.

Strengths:

(1) Strong and accessible technical innovation: With an elegant combination of beam shaping and optical modelling, the authors provide a high-resolution light-sheet system that overcomes the classical light-sheet tradeoff limit of a thin light-sheet and a small field of view. In addition, the integration of in silico modelling with a custom-machined baseplate is very practical and allows for ease of alignment procedures. Combining these features with the solid and super-extensive guide provided in the supplementary information, this provides a protocol for replicating the microscope in any other lab.

(2) Impeccable optical performance and ease of mounting of samples: The system takes advantage of the same sample-holding method seen already in other implementations, but reduces the optical complexity. At the same time, the authors claim to achieve similar lateral and axial resolution to Lattice-light-sheet microscopy (although without a direct comparison (see below in the "weaknesses" section). The optical characterization of the system is comprehensive and well-detailed. Additionally, the authors validate the system imaging sub-cellular structures in mammalian cells.

(3) Transparency and comprehensiveness of documentation and resources: A very detailed protocol provides detailed documentation about the setup, the optical modeling, and the total cost.

Weaknesses:

(1) Limited quantitative comparisons: Although some qualitative comparison with previously published systems (diSPIM, lattice light-sheet) is provided throughout the manuscript, some side-by-side comparison would be of great benefit for the manuscript,

> *even in the form of a theoretical simulation. While having a direct imaging comparison would be ideal, it's understandable that this goes beyond the interest of the paper; however, a table referencing image quality parameters (taken from the literature), such as signal-to-noise ratio, light-sheet thickness, and resolutions, would really enhance the features of the setup presented. Moreover, based also on the necessity for optical simplification, an additional comment on the importance/difference of dual objective/single objective light-sheet systems could really benefit the discussion.*

In the revised manuscript, we will expand our discussion to include a broader range of light-sheet microscope designs and imaging modes, including both single- and dual-objective configurations. We agree that highlighting the trade-offs between these approaches—such as working distance, sample geometry constraints, and alignment complexity—will enhance the overall context and utility of the manuscript.

To further aid comparison, we will include a summary table referencing key image quality parameters such as lateral and axial resolution, and illumination beam NA for Altair-LSFM. Where available, we will reference values from published work—such as the axial resolution reported in Valm et al. (Nature, 2017)—to provide a clearer benchmark. Because such comparisons can be technically nuanced, especially when comparing across systems with different geometries and sample mounting constraints, we will also include a supplementary note outlining the assumptions and limitations of these comparisons.

> *(2) Limitation to a fixed sample: In the manuscript, there is no mention of incubation temperature, CO_2 regulation, Humidity control, or possible integration of commercial environmental control systems. This is a major limitation for an imaging technique that owes its popularity to fast, volumetric, live-cell imaging of biological samples.*

We thank the reviewer for highlighting this important consideration. In the revised manuscript, we will provide a detailed description of how temperature control can be implemented using flexible adhesive heating elements, a power supply, and a PID controller. Step-by-step assembly instructions and recommended components will be included to facilitate adoption by users interested in live-cell imaging. We also note that most light-sheet microscopy systems capable of sub-cellular resolution—including the original LLSM design, diSPIM, and ASLM—typically do not incorporate integrated CO_2 or humidity control. These systems often rely on HEPES-buffered media to maintain pH stability, which is generally sufficient for short- to intermediate-term imaging. While full environmental control may be necessary for extended time-lapse studies, it is not a prerequisite for high-resolution volumetric imaging in many applications. Nonetheless, we will include a discussion of the challenges associated with adding CO_2 and humidity control to open or semi-enclosed architectures like Altair-LSFM, and outline potential future paths for integration with commercial incubation systems.

> *(3) System cost and data storage cost: While the system presented has the advantage of being open-source, it remains relatively expensive (considering the 150k without laser source and optical table, for example). The manuscript could benefit from a more direct comparison of the performance/cost ratio of existing systems, considering academic settings with budgets that most of the time would not allow for expensive architectures. Moreover, it would also be beneficial to discuss the adaptability of the system, in case a 30k objective could not be feasible. Will this system work with different optics (with the obvious limitations coming with the lower NA objective)? This could be an interesting point of discussion. Adaptability of the system in case of lower budgets or more cost-effective choices, depending on the needs.*

We thank the reviewer for raising this important point. First, we would like to clarify that the quoted $150k cost estimate includes the optical table and laser source. We apologize for any confusion and will communicate this more effectively in the revised manuscript.

We agree that adaptability is a key concern, especially in academic settings with limited budgets. The detection path can be readily altered depending on experimental needs and cost constraints. For example, in our discussion of alternatives to the 5 mm coverslip geometry, we will describe how switching to a Zeiss W Plan-Apochromat 20x/1.0 in combination with a compatible excitation objective allows high-resolution imaging while accommodating more conventional sample formats. We will expand this to include cost-effective alternatives as well.

We will also expand our discussion on cost-reduction strategies and the associated trade-offs. These include replacing motorized stages with manual ones, omitting the filter wheel in favor of a multi-band emission filter, or using industrial-grade cameras in place of scientific CMOS detectors. While each change entails some loss in functionality or sensitivity, such modifications allow users to tailor the system to their specific budget and application.

Finally, we recognize the challenge in communicating exact costs of commercial systems due to variability in configuration and pricing. Nonetheless, we will include approximate figures where possible and note that comparable commercial systems—such as LLSM platforms from 3i and Zeiss—are several-fold more expensive than the system presented here.

> *Last, not much is said about the need for data storage. Light-sheet microscopy's bottleneck is the creation of increasingly large datasets, and it could be beneficial to discuss more about the storage needs and the quantity of data generated.*

Data storage is indeed a critical consideration in light-sheet microscopy. In the revised manuscript, we will provide a note outlining typical volume dimensions for live-cell imaging experiments along with the associated data overhead. This will include estimates for voxel counts, bit depth, time-lapse acquisitions, and multi-channel datasets to help users anticipate storage needs. We will also briefly discuss strategies for managing large datasets, file types and compression formats.

> *Conclusion:*
>
> *Altair-LSFM represents a well-engineered and accessible light-sheet system that addresses a longstanding need for high-resolution, reproducible, and affordable sub-cellular light-sheet imaging. While some aspects-comparative benchmarking and validation, limitation for fixed samples-would benefit from further development, the manuscript makes a compelling case for Altair-LSFM as a valuable contribution to the open microscopy scientific community.*

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